Jenny Vo

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SCAN ME

EDUCATION

Olin College of Engineering – Needham, Massachusetts May 2028
Bachelor of Science : Electrical and Computer Engineering | GPA:4.0

- Relevant Coursework: Model Simulation, Quantitative Engineering Analysis, Software Design, Introduction to Sensors, Instrumentation and Measurement, Computer Architecture, Principles of Integrated Engineering

EXPERIENCE

Olin Baja – *Needham, Massachusetts* 2025-2026
Electrical Subteam

- Researched a GPS-based data acquisition system to capture race telemetry and generate accurate vehicle path traces for performance analysis.
- Integrated brake light and kill switch systems
- Performed rapid electrical troubleshooting and repairs under tight pit-stop time constraints during endurance competition, ensuring vehicle safety compliance and maximum uptime.

PROJECTS

32-Bit RISC-V Processor Core – *Computer Architecture, Olin College* Expected Dec 2026

- Designing an unpipelined, multicycle, 32-bit RISC-V processor with Harvard architecture.
- Implementing base instruction set to run a RISC-V assembly language program that blinks LEDs on a FPGA

CNC Sticker Maker and Drawing System | *GRBL, OpenJDK, Universal G-Code Sender, Arduino (C++)* *Fall* 2026

- Collaboration with a team of 5 to design and construct a multi-axis CNC sticker drawing and cutting system featuring automated drawer mechanics.
- Configured Arduino GRBL firmware with Universal G-Code Sender (UGS) for real-time G-code execution and motion control.

FPGA Game of Life & Color Cycling Matrix | *SystemVerilog, Python, iCE40 FPGA, Yosys, GTKWave* *Fall* 2026

- Designed a hardware implementation of Conway's Game of Life in SystemVerilog on an iCE40 FPGA, computing each generation by counting live neighbors per cell with modulo-based wrap-around indexing at the grid boundaries.
- Built a color-cycling system in a Python script generating RGB hue sequences as memory files and a hardware counter steps living cells along the color wheel on every generation.
- Verified grid transitions and next-state timing through Icarus Verilog simulation and GTKWave waveform analysis.

ZnO Nanorod Piezoelectric Energy Harvester | *Materials Chemistry, Olin College* Fall 2026

- Co-designed a flexible piezoelectric nanogenerator research proposal utilizing ZnO nanorod growth on ITO substrates.
- Researched and formulated hydrothermal growth protocols and PMMA polymer encapsulation techniques to optimize electrical response and mechanical durability under cyclic compression.

Automated 2D Pan/Tilt IR Distance Scanner | *Arduino, Sharp IR Sensor, Dual Servos, MATLAB* *Fall* 2026

- Engineered an automated two-axis infrared scanning mechanism driven by Arduino microcontrollers and dual servo actuators for precise polar orientation.
- Processed live spatial telemetry within MATLAB to render three-dimensional surface topology over a 20 cm to 150 cm operational radius.

Interactive River Bubble Game | *Python, OpenCV, MediaPipe, Pygame, PyTest* *Spring* 2026

- Worked with a team of 3 to build an interactive 2D game in Pygame controlled via live webcam feed using OpenCV and MediaPipe hand landmark tracking.
- Programmed real-time gesture control mapping open/closed hand combinations to lateral sprite movement across dynamic track obstacles.
- Integrated collision detection physics for environmental hazards and created unit test suites with PyTest to verify game state stability.

SKILLS

Programming Languages: MATLAB, Python(Pygame, Pytest, Pandas, OpenCV, MediaPipe), SystemVerilog, Arduino/C++
Software & Tools: OSS CAD Suite, Yosys, Iverilog (Icarus Verilog), GTKWave, KiCAD, SolidWorks, Microsoft Office
Hardware & Fabrication: Soldering, Crimping, Laser Cutting, 3D Printing

SCAN FOR PORTFOLIO

